Paper 1 — proposed revision of section 2.5

"Requiring at least k of N planes"

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Equation numbers are local to this note. References to other sections (2.2, 2.3, 2.6, 4) are to the current paper. The text below would replace the present section 2.5 entirely; notes for the authors follow at the end.

2.5 Requiring at least k of N planes

Why sets of planes. A real detector is not fully efficient, and a track finder that required a hit on every plane would lose every track with a missing hit. Track finders therefore accept candidates with hits on *at least* k of the N planes, five of six for example. Any such algorithm must also accept the noise combinations that pass on a subset of the planes. We therefore extend the counting of section 2.2 from one set of planes to all the sets that the algorithm accepts.

Candidates. Let S be a set of planes with $|S| \geq k$. A *candidate on* S is a choice of one hit on each plane of S, nothing being required of the planes outside S. It is fitted to the track model with the planes of S only, with $n_{\mathrm{dof}}(S) = 2|S| - 4$ degrees of freedom, and it is *accepted* if

$$\chi^2_S < c_S \quad \text{and} \quad \hat{\theta}_S \in \Theta, \tag{2.1}$$

where $\hat{\theta}_S$ are the fitted parameters. Each set has its own cut c_S, chosen at the same efficiency for real tracks as for the full set. With our one-sided 3σ choice the cut depends only on $n_{\mathrm{dof}}(S)$: for five of six planes, $c = 25.36$ for the full set ($n_{\mathrm{dof}} = 8$) and $c = 21.74$ for each of the six five-plane sets ($n_{\mathrm{dof}} = 6$).

The expected number of accepted candidates. We define $E(\geq k)$ as the expected number of accepted candidates, summed over all sets of at least k planes. A candidate on S is accepted with probability P_S, the probability of section 2.3 computed for the planes of S alone (their depths, sizes and resolutions), and there are $\prod_{i \in S} n_i$ candidates on S. By linearity of expectation, as in eq. (2.1),

$$E(\geq k) \;=\; \sum_{S:\,|S| \geq k} E_S, \qquad E_S \;=\; P_S \prod_{i \in S} n_i. \tag{2.2}$$

For at least five of six planes the sum has seven terms:

$$E(\geq 5) \;=\; P_{123456}\, n_1 n_2 n_3 n_4 n_5 n_6 \;+\; \sum_{j=1}^{6} P_{\bar{\jmath}} \prod_{i \neq j} n_i, \tag{2.3}$$

where $\bar{\jmath}$ denotes the five planes other than j. Note that a candidate on five planes is counted whether or not the sixth plane happens to have a hit near the track; this is what an algorithm accepting five of six planes does.

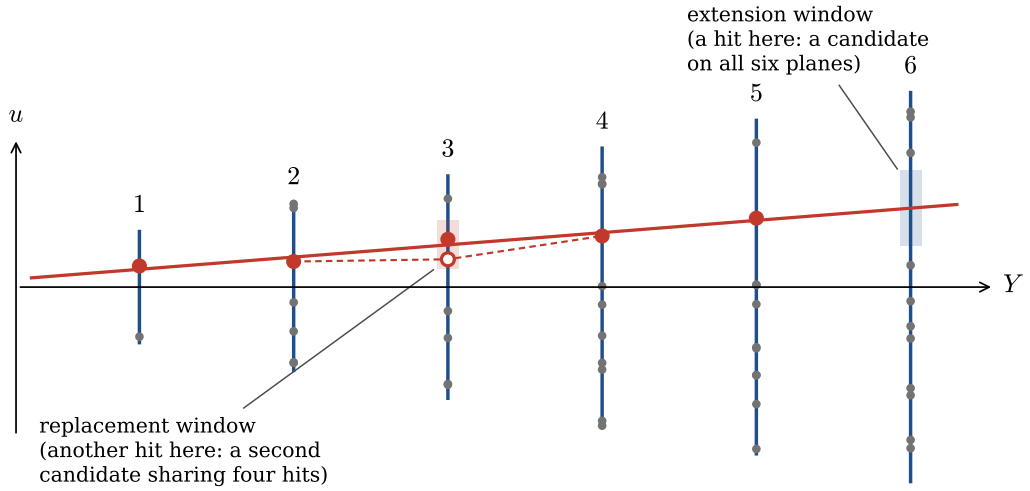


Figure 2.1: Candidates that share hits, shown in one view and not to scale. A candidate on planes 1–5 (filled red points) is accepted. The fit to planes 1, 2, 4 and 5 predicts where its hit on plane 3 must lie: any other hit in this *replacement window* (open point) forms a second accepted candidate on the same planes, sharing four hits with the first. The fit to planes 1–5 predicts plane 6: a hit in this *extension window* forms an accepted candidate on all six planes, which contains the first one.

Windows. Two accepted candidates can share hits, and the size of the overlap is governed by a single quantity, the number of hits expected in a *window* (figure 2.1). Take an accepted candidate on S and one of its planes, i. The fit to the other planes of S predicts the position $(\hat{u}_i, \hat{v}_i)$ of the track on plane i, with variances $V_{u,i}$ and $V_{v,i}$, and has its own $\chi^2_{S\setminus i}$. For a linear track model the χ^2 of the full set separates exactly into

$$\chi^2_S \;=\; \chi^2_{S\setminus i} \;+\; \frac{(u_i - \hat{u}_i)^2}{\sigma^2_{u,i} + V_{u,i}} \;+\; \frac{(v_i - \hat{v}_i)^2}{\sigma^2_{v,i} + V_{v,i}}. \tag{2.4}$$

Any hit (u, v) on plane i therefore forms an accepted candidate together with the other hits of S if it lies inside the ellipse in which the last two terms add up to less than $b_i = c_S - \chi^2_{S\setminus i}$, the part of the χ^2 budget not used by the other planes. The area of this ellipse is $\pi b_i \sqrt{(\sigma^2_{u,i} + V_{u,i})(\sigma^2_{v,i} + V_{v,i})}$.

The budget b_i varies from candidate to candidate, and its mean is simple. In the whitened space of section 2.3 the accepted noise combinations fill the n_{dof}-dimensional χ^2 sphere of radius $\sqrt{c_S}$ uniformly, and eq. (2.4) splits the squared distance from the centre into two coordinates for plane i and $n_{\mathrm{dof}} - 2$ for the rest. For a uniformly filled sphere each coordinate carries on average $c_S/(n_{\mathrm{dof}} + 2)$, so the mean of $\chi^2_{S\setminus i}$ is $c_S(n_{\mathrm{dof}} - 2)/(n_{\mathrm{dof}} + 2)$ and

$$\langle b_i \rangle \;=\; \frac{4\, c_S}{n_{\mathrm{dof}} + 2}, \tag{2.5}$$

which is $c_S/2$ for $n_{\mathrm{dof}} = 6$.

The window occupancy. The expected number of *other* noise hits in the replacement window of plane i is its area times the density of the other hits,

$$\eta_i \;=\; \frac{n_i - 1}{A_i}\, \pi\, \langle b_i \rangle \sqrt{(\sigma^2_{u,i} + V_{u,i})(\sigma^2_{v,i} + V_{v,i})}, \tag{2.6}$$

and the *window occupancy* of the candidate is the sum over its planes,

$$\eta \;=\; \sum_{i\in S} \eta_i. \tag{2.7}$$

It is the expected number of other accepted candidates on the same planes that share all but one hit with the given one. In the same way, for a plane j outside S crossed by the track, the expected number of hits in the extension window is

$$\mu_j \;=\; \frac{n_j}{A_j}\,\pi\,\langle b_j\rangle\,\sqrt{(\sigma_{u,j}^2+V_{u,j})(\sigma_{v,j}^2+V_{v,j})}, \qquad \langle b_j\rangle = c_{S\cup j} - \frac{n_{\mathrm{dof}}(S)}{n_{\mathrm{dof}}(S)+2}\,c_S, \tag{2.8}$$

with $V_{u,j}$, $V_{v,j}$ now the variances of the position predicted on plane j by the fit to all of S.

Why the sets with fewest planes dominate. Every accepted candidate on $S\cup j$ contains a candidate on S, and it is accepted when a hit falls in the extension window of plane j. Therefore, approximately,

$$E_{S\cup j} \;\simeq\; \mu_j\,E_S, \tag{2.9}$$

for the tracks of S that cross plane j. Each additional plane multiplies the expected number by the probability that it contains a compatible hit. When the windows are nearly empty, as they must be for tracking to work at all, $\mu_j \ll 1$ and the terms with exactly k planes dominate the sum (2.2), by a factor $1/\mu$. In the example of section 2.6, at 10^{-3} hits/mm^2, $\mu_6 = 6.0\times10^{-4}$ for the planes 1–5 and plane 6, and eq. (2.9) gives $E_{123456} \simeq 6.0\times10^{-4}\times 3.2\times10^{-6} = 1.9\times10^{-9}$, as computed directly from eq. (2.2).

From candidates to distinct fake tracks. $E(\geq k)$ counts candidates, and the same fake can appear as several of them, in two ways. (i) Two candidates on the same planes share all but one hit (the replacement window). (ii) A candidate on $S\cup j$ contains candidates on S (the extension window). A reconstruction algorithm reports such overlapping candidates as one track. We count as one *distinct fake* each group of accepted candidates connected by sharing at least $k-1$ hits; an algorithm cannot report fewer distinct fakes than groups without losing real tracks in the same configurations.

Overlaps of type (ii) are of relative size μ, already small whenever the sets of k planes dominate. For type (i), each accepted candidate has on average η partners. Each pair is seen from both of its candidates, so when overlaps are rare the number of groups is the number of candidates minus the number of pairs, $E(\geq k)(1-\eta/2)$. We find that

$$E_{\mathrm{distinct}} \;\simeq\; e^{-\eta/2}\,E(\geq k) \tag{2.10}$$

describes the counted number of distinct fakes within a few per cent up to $\eta \approx 2$ (section 4), with η averaged over the accepted candidates.

Summary. The lower limit on the number of distinct fake tracks for an algorithm that accepts at least k of N planes is E_{distinct} of eq. (2.10), with $E(\geq k)$ from eq. (2.2). In the regime where tracking is possible the windows are nearly empty, $\eta \ll 1$, and the two coincide: in the example of section 2.6 the five-plane candidates have $\eta \approx 3\times10^{-3}$, so E_{distinct} and $E(\geq 5)$ differ by 0.15%. Since η grows linearly with the density, it is still only about 0.02 at the density where one fake per event is expected (7.6×10^{-3} hits/mm^2), so the correction remains small wherever the limit matters.

Notes for the authors

1. **What is new compared with the present section 2.5.**

   - A definition of a candidate (a set of planes plus one hit on each) and of its acceptance, eq. (2.1), before $E(\geq k)$ is introduced; $E(\geq k)$ is then defined in words as well as by the formula, and written out term by term for five of six planes, eq. (2.3).
   - The windows are derived from the exact χ^2 split of eq. (2.4), with a figure, rather than stated.
   - The extension window and eq. (2.9) explain *why* the terms with the fewest planes dominate, with a numerical check against section 2.6.
   - The distinct-fake count is defined explicitly (groups sharing at least $k-1$ hits, which is what the validation of section 4 counts), and the origin of $1-\eta/2$ is explained.

2. **One correction.** The present eq. (2.8) uses $c/2$ for the mean budget. The exact mean for a uniformly filled χ^2 sphere is $4c/(n_{\mathrm{dof}}+2)$, eq. (2.5). It is $c/2$ for $n_{\mathrm{dof}}=6$, so nothing changes for the five-plane sets, which dominate; for the six-plane set ($n_{\mathrm{dof}}=8$) it is $0.4c$ instead of $0.5c$. The validation counts η hit by hit with the exact budget, so its results are unaffected.

3. **The lower-limit statement.** Strictly, the lower limit is on distinct fakes, $e^{-\eta/2}E(\geq k)$, not on $E(\geq k)$. The difference is negligible in practice (0.15% in the example), but section 2.2 and the conclusions should say "the expected number of distinct fakes" once the k-of-N case is in. I would add one sentence to section 2.2 pointing forward to this.

4. **Numbers.** The values $\eta \approx 3\times10^{-3}$ and $\mu_6 = 6.0\times10^{-4}$ are computed for the example of section 2.6 at 10^{-3} hits/mm^2 (all six five-plane sets have η between 3.0 and 3.3×10^{-3}). The check of eq. (2.9) works for the set 1–5 with plane 6 added. For sets missing an inner plane it needs the fraction of tracks that cross the missing plane (in the projective tower, many tracks through planes 2–6 miss the small plane 1), which is why the text says "for the tracks of S that cross plane j".

5. **Possible further simplification.** If the section still feels heavy, the extension-window paragraph (eqs. (2.8)–(2.9)) could move to the numerical example, keeping here only the replacement window and η. I would keep it here: it is the shortest explanation of why five-plane sets dominate, which is the most striking result of the example.

6. **The figure** is a schematic in one view, deliberately not to scale (real windows are a few resolutions wide on planes hundreds of millimetres across). It could go in the paper as is, or be merged with figure 1 as a second panel.